Chapter 1: Solutions

1.1 Introduction

A solution is a homogeneous mixture of two or more components. The component present in the largest quantity is called the solvent. The component present in smaller quantity is called the solute. A binary solution has two components: one solute and one solvent.

Solutions can be classified by the physical state of the solvent into solid, liquid, and gaseous solutions. The most common type in chemistry is the liquid solution where the solvent is a liquid.

1.2 Types of Solutions

Definition: A saturated solution is a solution in which no more solute can dissolve at a given temperature.

Definition: An unsaturated solution is a solution in which more solute can be dissolved at the same temperature.

Definition: A supersaturated solution is an unstable solution containing more solute than the saturation amount at the given temperature.

1.3 Expressing Concentration of Solutions

Several methods are used to express the concentration of a solution.

Definition: Mass percentage of a component in a solution is the mass of that component expressed as a percentage of the total mass of the solution.

Mass percentage of A equals the mass of A divided by total mass of solution, times one hundred.

Definition: Molarity (M) is the number of moles of solute dissolved per litre of solution. It depends on temperature because volume changes with temperature.

Formula: M equals moles of solute divided by volume of solution in litres.

Definition: Molality (m) is the number of moles of solute per kilogram of solvent. Molality is temperature-independent because mass does not change with temperature.

Formula: m equals moles of solute divided by mass of solvent in kilograms.

Definition: Mole fraction of a component is the number of moles of that component divided by the total number of moles of all components in the solution.

Example 1.1: Calculate the molality of a solution containing 4.0 grams of NaOH dissolved in 250 grams of water. Molar mass of NaOH is 40 g per mol.
Solution: moles of NaOH equals 4.0 divided by 40 equals 0.1 mol. Mass of solvent in kg equals 0.250. Molality equals 0.1 divided by 0.250 equals 0.4 mol per kg.

1.4 Solubility

Definition: Solubility of a substance is the maximum amount that can be dissolved in a specified amount of solvent at a specified temperature.

Solubility of a solid in a liquid depends on the nature of the solute and solvent, on temperature, and weakly on pressure.

Henry's Law: The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution. Mathematically, p equals K H times x, where K H is the Henry's law constant.

Example 1.2: If N2 gas is bubbled through water at 293 K and 1 atm, calculate the mole fraction of N2 in water. K H for N2 at 293 K is 76.48 kbar.
Solution: x equals p divided by K H equals 1 atm divided by 76.48 kbar. Converting 1 atm to 0.987 bar, x equals 0.987 divided by 76480 equals 1.29 times 10 to the minus 5.

1.5 Vapour Pressure of Liquid Solutions

Raoult's Law for a solution of two volatile components: the partial vapour pressure of each component is directly proportional to its mole fraction in the solution. p1 equals p1-naught times x1, and p2 equals p2-naught times x2, where p1-naught and p2-naught are vapour pressures of pure components.

1.6 Colligative Properties

Definition: Colligative properties are properties that depend only on the number of solute particles, not on their nature.

The four colligative properties are: relative lowering of vapour pressure, elevation of boiling point, depression of freezing point, and osmotic pressure.

Elevation of boiling point: delta T b equals K b times m, where K b is the ebullioscopic constant and m is the molality of the solute.

Depression of freezing point: delta T f equals K f times m, where K f is the cryoscopic constant.

Osmotic pressure: pi equals C R T, where C is the molar concentration of the solute, R is the gas constant, and T is the temperature in Kelvin.

Example 1.3: 18 grams of glucose (molar mass 180) is dissolved in 1 kg of water. Calculate the freezing point depression. K f for water is 1.86 K kg per mol. Solution: molality m equals 18 divided by 180 divided by 1 equals 0.1 mol per kg. Delta T f equals 1.86 times 0.1 equals 0.186 K. So the freezing point of the solution is minus 0.186 degrees Celsius.

1.7 Abnormal Molar Masses

When the actual molar mass differs from the expected value, the solute is said to show abnormal behaviour due to association or dissociation. The van't Hoff factor i corrects for this. i equals normal molar mass divided by abnormal molar mass.

For complete dissociation of an electrolyte AB into A plus and B minus, i approaches 2. For dimerization of an acid like benzoic acid in benzene, i approaches 0.5.